Why SAMURAI73 Needs Two Polarimeters

Full tensor reconstruction and locating p_{yy} loss

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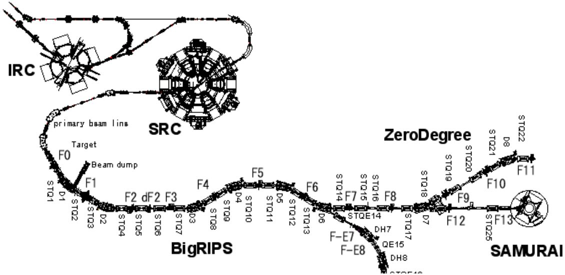
IVR needs the tensor state delivered to the target



First IVR run

Start with vertical p_{yy}; longitudinal p_{zz} is harder to prepare.

After SRC

What polarization leaves the accelerator chain?

After BigRIPS

What polarization reaches SAMURAI73?

Two stations are needed to determine all degrees of freedom of the spin-density matrix.

Spin-1: 8 general degrees, 5 tensor, 1 prepared

For $w_a \geq 0$ and $\sum_a w_a = 1$,

$$\rho = \sum_a w_a |\psi_a\rangle\langle\psi_a|.$$

A normalized 3×3 Hermitian ρ has

$$\boxed{8 = 3_{\text{vector}} + 5_{\text{tensor}}}$$

independent real parameters.

An axial tensor prepared about $\hat{\boldsymbol{n}}$ is

$$\boldsymbol{P}_0^{(2)}(q, \hat{\boldsymbol{n}}) = \frac{q}{2}\left(3\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^{\mathsf{T}} - I\right).$$

$$\hat{\boldsymbol{n}} = \hat{\boldsymbol{y}}: \quad \boldsymbol{P}_0^{(2)} = \frac{p_{yy}}{2}\,\mathrm{diag}(-1, 2, -1),$$

$$\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}}: \quad \boldsymbol{P}_0^{(2)} = \frac{p_{zz}}{2}\,\mathrm{diag}(-1, -1, 2).$$

Tensor only: 5 degrees. Fixed preparation axis: one amplitude q.

Linear BMT preserves vector and tensor ranks

1. Magnetic field rotates the spin in the lab

$$\rho_{\mathrm{lab},f} = U_s \rho_0 U_s^\dagger, \qquad R_s \in SO(3).$$

Thomas–BMT determines R_s along the trajectory.

2. The downstream beam defines new axes

$$\rho_{b,f} = U_o^\dagger \rho_{\mathrm{lab},f} U_o.$$

R_o follows the transported momentum and defines the local x_b, y_b, z_b frame.

$$\boxed{R_{\mathrm{rel}} = R_o^{\mathsf{T}} R_s}, \qquad \boldsymbol{p}_{b,f} = R_{\mathrm{rel}} \boldsymbol{p}_0, \qquad \boldsymbol{P}_{b,f}^{(2)} = R_{\mathrm{rel}} \boldsymbol{P}_0^{(2)} R_{\mathrm{rel}}^{\mathsf{T}}.$$

Lowest order: a common rotation moves the polarization axis, but vector and tensor ranks remain separate.

BMT field spread cannot convert tensor to vector

Each phase-space point $\xi = (x, x', y, y', \Delta p/p, \ldots)$ has its own calculated rotation:

$$\boldsymbol{p}_f(\xi) = R_\xi \boldsymbol{p}_0, \qquad \boldsymbol{P}_f^{(2)}(\xi) = R_\xi \boldsymbol{P}_0^{(2)} R_\xi^{\mathrm{T}}.$$

The polarimeter measures the ensemble average:

$$\overline{\boldsymbol{p}}_f = \int d\xi\, f(\xi) R_\xi \boldsymbol{p}_0, \qquad \overline{\boldsymbol{P}^{(2)}}_f = \int d\xi\, f(\xi) R_\xi \boldsymbol{P}_0^{(2)} R_\xi^{\mathrm{T}}.$$

If $\boldsymbol{p}_0 = 0$, then exactly

$$\overline{\boldsymbol{p}}_f = 0.$$

The averaged tensor can become biaxial and require all

$$\boxed{\text{5 tensor components}}.$$

Different $B(\xi)$ under linear BMT can open up to 5 tensor degrees; it cannot give tensor→vector conversion.

Rank mixing opens all 8 state parameters

Tensor–vector conversion requires spin dynamics beyond the linear BMT term. For example,

$$H_{\rm spin} = -\mu\, \boldsymbol{S}\cdot\boldsymbol{B} - \beta_T \left(\boldsymbol{S}\cdot\boldsymbol{B}\right)^2 + \cdots, \qquad \overline{\rho}_f = \int d\xi\, f(\xi)\, \mathcal{E}_\xi[\rho_0].$$

Quadratic spin interaction

Tensor magnetic/electric polarizability or other quadratic spin couplings can generate

$$\boldsymbol{p}_0 = 0,\ \boldsymbol{P}_0^{(2)} \neq 0 \quad \longrightarrow \quad \boldsymbol{p}_f \neq 0.$$

Spin-dependent loss or material interaction

Spin-dependent loss, material interactions, collimation, and acceptance can change both ranks and the density-matrix eigenvalues.

Once rank mixing is allowed, the fit must allow the full $8 = 3_{\rm vector} + 5_{\rm tensor}$ parameters.

The cross section shows the two blind directions

$$
\begin{aligned}
\frac{\sigma(\theta,\phi)}{\sigma_0(\theta)} &= 1 + \frac{3}{2}\,(p_x \sin\phi + p_y \cos\phi)\, A_y(\theta) \\
&+ \frac{2}{3}\,(p_{xz} \cos\phi - p_{yz} \sin\phi)\, A_{xz}(\theta) \\
&+ \frac{1}{6}\,[(p_{xx} - p_{yy}) \cos 2\phi - 2p_{xy} \sin 2\phi]\,[A_{xx}(\theta) - A_{yy}(\theta)] \\
&+ \frac{1}{2} p_{zz} A_{zz}(\theta).
\end{aligned}
$$

p_z is absent

The d–p reaction has no longitudinal-vector term in this geometry.

p_{xy} vanishes at the four arms

At L/R/U/D, $\phi = 0^\circ, 90^\circ, 180^\circ, 270^\circ$, so $\sin 2\phi = 0$.

One calibrated multi-angle station measures 6/8 density-matrix parameters, or 4/5 tensor parameters.

BigRIPS makes both blind directions visible

Before BigRIPS: unprimed. **After BigRIPS:** primed.

$$R_{\mathrm{rel}} = R_y(\delta), \qquad \delta = 27.513^\circ, \qquad c = \cos\delta = 0.886906, \qquad s = \sin\delta = 0.461951.$$

Vector blind direction

$$p'_x = c\,p_x + s\,p_z,$$
$$p'_z = -s\,p_x + c\,p_z.$$

Hence an upstream p_z gives

$$\boxed{p'_x = 0.461951\,p_z},$$

which is measured at station 2.

Tensor blind direction

$$p'_{xy} = c\,p_{xy} + s\,p_{yz},$$
$$p'_{yz} = -s\,p_{xy} + c\,p_{yz}.$$

Hence an upstream p_{xy} gives

$$\boxed{p'_{yz} = -0.461951\,p_{xy}},$$

which is measured at station 2.

Two stations: tensor $5/5$; full density matrix $8/8$.

Why can p_{yy} decrease?

Compare tensor size Q_T, vector polarization $V = |\boldsymbol{p}|$, and density-matrix purity $\Pi = \mathrm{Tr}(\rho^2)$.

what the polarimeters see	what happened in the beam
p_{yy} **changes;** Q_T **and** Π **stay constant; the axis moves;** $V = 0$	Common magnetic rotation: all particles' axes turn together; no polarization is lost.
Q_T **changes;** V **appears;** Π **stays constant**	Quadratic spin coupling transfers polarization from tensor to vector.
Π **decreases**	Different trajectories acquire different axes or phases and cancel in the average; spin-dependent loss can also change ρ.

Changed already after SRC $\Rightarrow$ upstream. Changed only after BigRIPS $\Rightarrow$ between the stations.

Who should we contact to start the after-SRC process?

We would like to start the formal installation process.

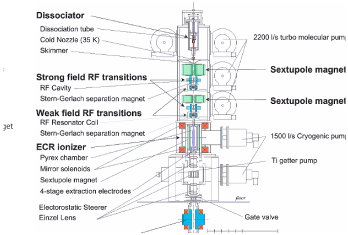
1. Who owns the approval for an installation immediately downstream of SRC?
2. What is the required technical, accelerator, safety, and scheduling sequence?
3. Where should we submit the first CAD/interface package, and what documents are required?

Requested outcome: one named contact and the next formal step.

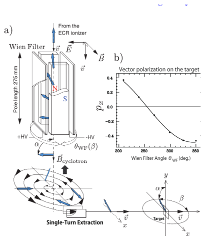
Backup

Source, accelerator, beam-line, and detector details

Backup: Wien filter sets the PIS spin axis



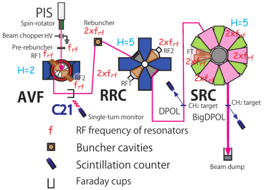
RIKEN polarized ion source



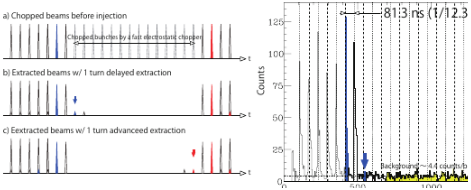
Wien-filter spin-orientation control

- The two-stage PIS produces vector and tensor modes.
- After the ECR ionizer the axis is vertical; crossed $\boldsymbol{E}$ and $\boldsymbol{B}$ tilt the spin without changing the orbit.
- Rotating the Wien filter sets the azimuth; vector polarization calibrates the angles.

Backup: single-turn extraction preserves spin phase



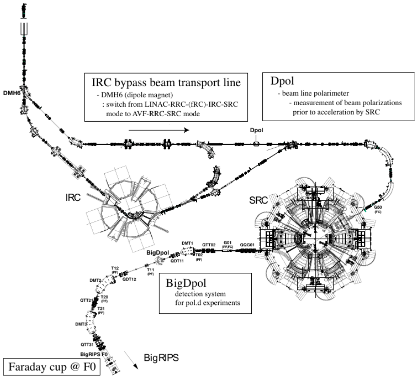
PIS $\rightarrow$ AVF $\rightarrow$ RRC $\rightarrow$ SRC

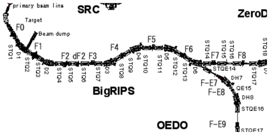


Timing signature of wrong-turn extraction

- The cyclotron field changes the spin direction every turn: $\Delta\beta = 360^\circ\gamma(g-1)$, with $g = 0.8477$ in the paper's convention.
- Mixing N with $N \pm 1$ turns superposes spin phases and lowers the polarization.
- AVF, RRC, and SRC kept the same turn: 0.07% mixing, $> 99\%$ purity, stable for 2–3 days.

Backup: magnets between the two stations





- Station 1: BigDpol/T11, between DMT1 and DMT2.
- DMT2 and DMT3: two identical 50° dipoles.
- BigRIPS $F0 \to F7$: six 30° dipoles, signs $(-, -, -, +, +, -)$, net bend $-60°$.
- $F7 \to F13$: quadrupoles, but no incident-beam dipole before the target.

Backup: central ray gives $\delta = 27.513^\circ$ at 190 MeV/u

DMT2 + DMT3	-100°
BigRIPS $F0 \to F7$	-60°
$F7 \to F13$	0°
total orbit bend θ	-160°

For a 190 MeV/u deuteron,

$$\gamma = 1.20260044, \qquad G_d = -0.14298727,$$

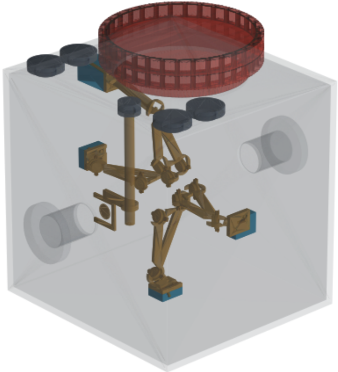
$$\delta = G_d\gamma\theta = +27.513^\circ, \qquad R_{\mathrm{rel}} = R_y(\delta).$$

The central rotation exposes the one-station blind directions:

$$p'_x = 0.461951\, p_z + \cdots, \qquad p'_{yz} = -0.461951\, p_{xy} + \cdots.$$

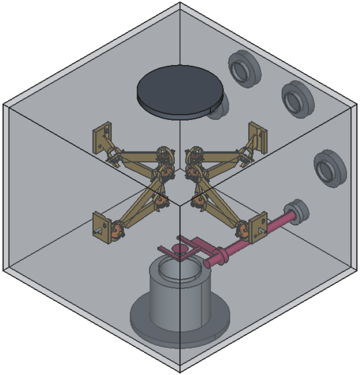
This is the central-ray horizontal-bend baseline; the real analysis uses $R_{\mathrm{rel}}(\xi)$ from field maps and measured beam phase space.

Backup: proposed polarimeter after SRC



Compact in-vacuum polarimeter proposed just downstream of SRC.

Backup: proposed polarimeter before SAMURAI



Compact in-vacuum polarimeter proposed upstream of SAMURAI.

Backup: expected p_{yy} precision and rates

Detector geometry

channel	θ_{lab}	radius
deuteron	20.9°	140 mm
proton, small angle	11.2°	190 mm
proton, large angle	53.4°	205 mm

Rate assumptions

beam	190 MeV/u, 10^7 d/s
nominal polarization	$p_{yy} = 0.8$
CH_2 for rate model	$1000\,\mathrm{mg/cm^2}$
selected LR/UD rate	$36.95\,\mathrm{s^{-1}}$

time	$N_L + N_R$	$N_U + N_D$	total	p_{yy} statistical precision
10 min	13,734	8,438	22,172	$\sigma = 0.0210$; 68% CI $[0.779, 0.821]$
30 min	41,203	25,314	66,517	$\sigma = 0.0121$; 68% CI $[0.788, 0.812]$

Counts use $1000\,\mathrm{mg/cm^2}$, not the 0.20 mm target drawn in the CAD.